

Belief State: A Graph Architecture for Narrative Consistency in LLM-Driven NPCs

Master's Thesis Summary

Candidate: Andrea Lecini

Degree Course: Cinema and Media Engineering

Advisors: Marco Mazzaglia, Stefano Calzolari

Institution: Politecnico di Torino (A.Y. 2025/2026)

1. PROBLEM ADDRESSED

Integrating Large Language Models (LLMs) into role-playing games removes the structural limitations of traditional dialogue trees—the closed conversational space—while eliminating their implicit virtue: because that space was authored, it was consistent by design. In its place appears a less visible and more insidious flaw, which this thesis terms **narrative panic switching**: non-player characters (NPCs) contradict, without any narrative justification, what they previously stated or promised. This is not a reasoned change of opinion driven by events, but an uncaused leap: an NPC who promised to wait denies having made the promise; a betrayed NPC welcomes the betrayer as if nothing happened.

The thesis identifies three distinct causes of this phenomenon: *context window truncation*, *lossy summarization*, and the *absence of an explicit belief state*. It argues that intervention must target the third cause because it renders the other two damaging: as long as binding information relies on conversational text, truncating or summarizing it destroys that information. The observable effect is selective narrative amnesia: major events survive, while micro-commitments and relational details—the very elements that build a character's sense of identity—disappear, as summarization preserves what is salient rather than what is logically binding.

Research Question: Can an explicit belief state management system reduce narrative inconsistencies in LLM-driven NPCs, and how can an evaluation metric for temporal consistency in extended gameplay sessions be formalized?

The research question is deliberately twofold (design and methodology). From this arise three operational objectives: (1) design a structured, persistent belief registry; (2) resolve conflicts deterministically prior to model calls; (3) quantitatively measure consistency over extended gameplay sessions.

2. POSITIONING IN THE STATE OF THE ART

Chapter 2 categorizes existing research across three domains: agent memory (*Generative Agents*, *Voyager*, *PokéLLMon*; graph memories like *AriGraph* and *Zep/Graphiti*), belief representation (*ABBEL*, *BEE*, *Belief Engine*), and conflict detection with metrics (*STALE*, *SAVER*, *Kumiho* with AGM semantics, *BeliefShift*). A comparative analysis across seven criteria reveals three major research gaps:

- **Cascading Propagation:** No existing work fully covers cascading conflict resolution. *STALE* taxonomizes it without solving it; *Kumiho* acknowledges it; others ignore it. Yet, propagation distinguishes a true belief registry from a static list of facts.
- **Pre-generation Detector + Post-generation Verifier:** Existing architectures test either before or after generation, but never both, missing distinct failure modes.
- **Narrative Metric:** Existing metrics track drift or explainability, but none measure an NPC's character consistency throughout an active gameplay session.

3. DOMAIN ANALYSIS: RELATIONAL INTERACTION MECHANICS (MIR)

Before introducing the software architecture, the work anchors its contribution within game design. Persuasion, bribery, intimidation, trust-building, and betrayal are formalized as parametric variations of a single core mechanic defined by the quadruple $MIR = (S, M, T, O)$: relational state space S , communicative move space M , transition function $T: S \times M \rightarrow S$, and observation function $O: S \rightarrow B$.

The core theoretical diagnosis reveals that while LLM integration expands *Procedures* and *Resources* (Fullerton's framework), the *Rules* layer remains flawed: the transition function T collapses all observations into a single state update, permanently losing the underlying cause. For example, obedience out of loyalty and obedience out of fear are observationally identical in the present, but predict opposite behaviors over time. LLMs do not eliminate this state collapse; they merely render it less inspectable.

Base Architecture & Latency Budget

The implementation is built upon **Mantella**, a mod for *The Elder Scrolls V: Skyrim* connecting the game engine to an LLM and a text-to-speech (TTS) pipeline. Mantella's long-term memory relies on a textual summary file per character—the exact point where consistency-critical details are lost in a self-reinforcing error loop. Instrumentation of the baseline pipeline showed: input acquisition (0.0006s), prompt construction (0.177s), TTFT (0.590s), and full response streaming (1.346s). Including voice synthesis, total perceived delay reaches 4–5 seconds—establishing a strict budget within which any architectural extension must operate.

4. CONTRIBUTION: THE BELIEF STATE NODE

The central design principle is strict: **a character's relational state is not reconstructed by the language model from dialogue text, but maintained explicitly outside of it and resolved for consistency before generation takes place.** The LLM is tasked only with expressing an already resolved state in natural language.

Developed in Python and integrated under `src/beliefstate`, the system consists of four key modules:

- Claim Extraction:** Translates available sources into propositions conforming to a closed schema. Sources include SKSE game state (`source_type = ENGINE`, highest trust) and player dialogue (`PLAYER_DIALOGUE`).
- Belief State Node:** A persistent Directed Acyclic Graph (DAG) per character. Each node represents a typed proposition (*Fact*, *Intention*, *RelationalState*, *Commitment*) with status (`ACTIVE/SUPERSEDED`), `source_type`, and Skyrim calendar timestamps. The sole edge type is `depends_on`, used for derived cascade rules.
- Contradiction Detector:** Evaluated on every insertion before generation. Detects Type I conflicts (direct key collisions with differing values) and Type II conflicts (cascading conflicts propagated via dependency rules with inverted roles).
- Action Verifier:** Post-generation verification that extracts claims from LLM responses and validates them against the graph before sending output to voice synthesis.

Conflict Resolution Policy (Source Hierarchy):
When a conflict is detected, the proposition with the higher-tier `source_type` wins. Recency breaks ties only within the same tier:

$$ENGINE > DERIVED_CASCADE > PLAYER_DIALOGUE > LLM_GENERATED$$

This priority prevents observation collapse: if a player claims they never possessed an item that the game engine reports in their inventory, the engine fact overrides the player's statement.

Prompt Disambiguation

Replacing verbose text summaries with compressed proposition blocks eliminates semantic ambiguity. Stating `trust(Agnis, Player) = low` leaves no room for dual interpretation, removing the cognitive inference step from the LLM.

Direct Memory Logging (DML)

Abstracts a direct memory channel between the game environment and agent memory (via State-Snapshot Buffer and Episodic Event Log), bypassing perception errors for structured symbolic data.

5. EXPERIMENTAL DESIGN AND METRICS

All experiments evaluated two setups under identical conditions: **Baseline** (unmodified Mantella) vs. **Proposed** (with Belief State Node). Evaluation was carried out inside a custom-built Skyrim cell featuring isolated test quests and duplicated NPCs.

Metric	Definition	Target Measurement
CRR	Resolved Conflicts / Total Conflicts (Direct & Cascade)	Accuracy of the resolution engine; paired with false positive rate on distractor scenarios.
AVOR	Discarded Responses / Total Generated Responses	Frequency of the model attempting inconsistency despite receiving valid context.
NCS	Consistent Touchpoints / Verifiable Touchpoints	Observed narrative consistency across active narrative checkpoints.
Latency	Mean Turn Overhead (Proposed –	Real-world computational practicality (directly measured).

6. RESULTS

Experiment A – Injected Conflict Benchmark

Evaluated performance across 4 scenario templates (direct conflict, cascade, commitment, distractor) repeated 5–7 times across three extraction models.

Condition	Direct Conflicts	Cascading Conflicts	Total CRR
Baseline (Mantella)	15 / 16 (94%)	22 / 32 (69%)	37 / 48 (77%)
Belief State (qwen2.5:7b)	16 / 16 (100%)	32 / 32 (100%)	48 / 48 (100%)

The gap is concentrated in cascading conflicts (23% difference), where unstated inferences are required. Direct conflicts are handled relatively well by baseline LLMs when contained within short context windows.

Extraction Model	Direct CRR	Cascade CRR	Total CRR	Key Observations
qwen2.5:3b	18 / 18 (100%)	16 / 32 (50%)	34 / 50 (68%)	Fast, but produces extraction direction errors & false positives.
qwen2.5:7b	16 / 16 (100%)	32 / 32 (100%)	48 / 48 (100%)	Optimal balance of latency and reliability; zero logic errors.
gpt-oss:120b	15 / 15 (100%)	31 / 31 (100%)	46 / 46 (100%)	Highly robust to vague input, but excessively slow.

Experiment B – Long-Horizon Narrative Consistency

Simulated extended play (~100 turns across 3 sessions; Agnis/betrayal and Aela/broken commitment) using an LLM-as-Player agent with distinct seeds.

Condition	Inconsistent Responses	Narrative Consistency Score (NCS)
Baseline (Mantella)	21.0% (observed in text)	63.0%
Belief State Node	0.8% (AVOR overrides)	100.0%

An Action Verifier override rate of only 0.8% indicates that when provided with a resolved context, the LLM almost never generates inconsistent statements. The Verifier acts as a safety net rather than a constant filter. Baseline consistency degraded steadily over time as episodic memory grew and was summarized.

Experiment C – Human Participant Study

20 participants, 40 gameplay sessions, 127 responses evaluated in a counterbalanced 8-phase trial with Lydia (3 narrative touchpoints, 23-item questionnaire based on the Guo-Adamo-Mousas credibility scale).

Measured Outcome	Baseline	Belief State	Statistical Test & Significance
Observed NCS (Execution Logs)	40 / 60 (66.7%)	57 / 60 (95.0%)	Significant improvement in log validation
Perceived NCS (User Survey)	42 / 60 (70.0%)	58 / 60 (96.7%)	Wilcoxon $W = 13.0$, $p = 0.0017$
Dichotomized Touchpoints	2 in favor	20 in favor	Exact McNemar test, $p = 0.00012$
Perceived Character Credibility	4 / 20	16 / 20	Binomial test, $p = 0.012$
Memory Accuracy Assessment	4 / 20	16 / 20	Binomial test, $p = 0.012$
Conversational Fluidity	15 / 20	5 / 20	Baseline favored due to lower latency
Willingness to Continue Dialogue	9 / 20	11 / 20	$p = 0.824$ (Not statistically significant)

Latency Overhead Analysis

Mean response latency (player input to speech start) increased from **4.87s (baseline)** to **9.72s (proposed)**—adding ~5 seconds per turn. In user studies, median turn processing delay rose from 1.48s to 7.07s (+4.96s to +6.00s stable delta across participants).

Two key insights bound this result: (1) latency is concentrated entirely in claim extraction calls (2.72s and 3.06s average), whereas the graph logic execution takes negligible time; (2) downstream verification is currently executed synchronously on all turns, despite triggering an override on only 0.8% of simulated turns (8.7% in user sessions).

7. SUSTAINED THESIS, LIMITATIONS, AND FUTURE WORK

Core Thesis: Narrative consistency does not emerge spontaneously from language models; it requires structural architectural constraints.

This thesis is supported by four main observations:

1. The same generative model improves from 77% to 100% resolution accuracy and from 63% to 100% narrative consistency when constrained by explicit graph architecture.
2. A 7B parameter model paired with a deterministic graph engine achieves complete consistency, equaling a model 17 times larger.
3. Baseline failures cluster around cascading conflicts, proving that missing information—not model capacity—causes narrative failure.
4. Consistency gains are recognized by human players in controlled empirical trials.

Declared Limitations

- **Extraction Bottleneck:** Synchronous claim extraction causes noticeable latency penalties.
- **Closed Registries:** Predicates and entities rely on fixed schemas, limiting open-ended world interaction.
- **Isolated Character Graphs:** Graphs are maintained per character, excluding multi-agent belief exchange or theory-of-mind modeling.
- **Sample Scope:** User validation (20 users, 12-minute sessions) serves as directional support rather than exhaustive validation.

Future Work Directions

- **Latency Optimization:** Asynchronous downstream verification and lightweight specialized models for claim extraction.
- **Social Graph Expansion:** Multi-agent belief propagation to model dynamic social networks and dynamic gossip.
- **Multimodal Integration:** Connecting explicit relational states to non-verbal expressivity (gestures, facial expressions).